

Aryan Nath

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Education

University of California, San Diego

Bachelor of Science

San Diego, CA, USA

Sep 2024 - Expected March 2027

- **Major:** Data Science, **Track:** Machine Learning and Artificial Intelligence
- Provost Honors; **GPA:** 3.84/4.0
- Relevant Coursework: Machine Learning, Data Structures & Algorithms, Probability and Statistics, Time Series Analysis

Experience

AgentSpace AI Researcher/Developer

October 2025 - Present

Computer Science and Engineering Society, UCSD

- Developed LangGraph **Graph-RAG system components** for an agentic AI platform simulating expert decision-making, designing schemas for precise, **context-aware** retrieval.
- **Implemented sequential decision logic** using **Reinforcement Learning** and game-theoretic models to calibrate agent personalities and optimize user engagement.

Automation & Game Systems Developer

August 2025 - Present

USP Digital

- **Designed, built, and shipped** a Unity sorting game with 5+ categories, integrating audio-visual effects and responsive UI to maximize player engagement.
- Created automated asset workflows via custom **Unity editor C#** tools, accelerating release cycles; resolved **30+** critical bugs to ensure consistent user experience.

Quantitative Analyst

Mar 2025 - Present

Sustainable Investment Group, UCSD

- Modeled equity volatility using **ARIMA** and **GARCH**; integrated ESG materiality into valuation drivers and presented a risk-adjusted **buy-sell thesis** to an industry panel.
- Engineered a market-neutral strategy by clustering 477 stocks via **PCA** and **t-SNE** on CAPM residuals; defined 11 distinct behavioral groups for analysis using **K-Means clustering**.
- Identified **27** statistically significant pairs for predictive trading; backtested strategy performance using **Sharpe Ratio, Sortino Ratio, and Maximum Drawdown** to verify risk-adjusted returns.

Projects

BioSync - Bio-informed Habit Development

June 2025 - Present

- Working on the design and development of BioSync, a data system to **optimize habits** through real-time physiological and environmental data integration from wearable devices.
- Designing a Python ETL pipeline to ingest and synchronize **multimodal wearable data** (HRV, SpO2, sleep).
- Training **LSTM/TCN** models for state forecasting and XGBoost/Random Forest for classification, tracking MAE and RMSE across releases.

Tennis Match Outcome Prediction

September 2025

- Designed a **scalable PostgreSQL** database and **automated ETL pipelines** to integrate 74,000+ historical ATP matches (2000-2024) and high-resolution NOAA environmental data.
- Optimized Random Forest and Logistic Regression classifiers with **stratified K-fold** validation, achieving **>70%** accuracy validated by strong ROC-AUC scores.

Playlist to Podium - Datahacks 2025

April 2025

- Engineered Python pipelines to process **1.7M+** audio features from Spotify API/Kaggle, enabling **scalable similarity scoring** for personalized music profiles.
- Simulated F1 season outcomes using **multiple regression** models on playlist inputs.

Awards & Recognition

Halicioğlu Data Science Institute Undergraduate Research Scholarship

June 2025 - June 2026

- Awarded \$2,500 for **independent** development of BioSync, an identified high-impact Data Science project selected from ~300 applicants (<3% acceptance rate).

Skills

Languages/Backend: Python, Java, PostgreSQL, C#, Unity, MERN (MongoDB, Express, React, Node.js)

ML/Statistics: NumPy, Pandas, scikit-learn, PyTorch, XGBoost, Random Forests, Time Series (ARIMA), LSTM/TCN, Reinforcement Learning (Contextual Bandits)